

Interpreting FCFS Appointment Simulation Results Under the Current Formulation

Metric behavior, access losses, and model checks

CUIMC Appointment Simulation

May 2026

Contents

1 Abstract

This note interprets simulation outputs from a first-contact FCFS appointment model. It is an *interpretation note*, not an empirical claim about real clinic operations. All statements are conditional on the current model formulation, and numbers should be read as model outputs, not clinical measurements.

Main finding. Under the current formulation, average utilization and the overall served rate separate sharply. At baseline, average utilization is **0.839** while the overall served rate is **0.269**. This gap is mechanically plausible rather than paradoxical: utilization is normalized by the number of available slots, while the served rate is normalized by the number of patient arrivals. With 100 arrivals per day and 32 slots per day the system is structurally oversubscribed. Even if every scheduled slot were completed without any no-show, only $32/100 = 0.32$ of arrivals could be served. The observed served rate of 0.269 is further reduced below this ceiling by no-shows.

Baseline loss decomposition. Of all arrivals at baseline, approximately 0.356 balk (reject the offered appointment), 0.323 cancel after booking, and 0.052 are no-shows. No-offer (horizon-full rejection) is zero at baseline, confirming that the booking horizon is not saturated under baseline demand. The low served rate is therefore driven primarily by balking and post-booking cancellation, not by horizon saturation.

Offered vs. accepted wait. Mean offered wait (9.30 days at baseline) includes both patients who accepted and patients who later balked. Mean accepted wait (8.35 days) excludes balked patients and is therefore a selected, downward-biased measure of the delay experience. Offered wait should always be reported alongside the no-offer share, because patients who receive no offer are excluded from both wait metrics.

2 Simulation Mechanics Relevant for Interpretation

The points below are necessary for reading the metric outputs correctly. They describe the current implementation rather than design aspirations.

- **Finite booking horizon.** The simulation maintains a rolling calendar of $H = 14$ residual days. When all days are fully booked, the next arrival receives no offer.
- **Two patient classes.** Each class has its own Poisson arrival rate, balking rule, no-show rule, and cancellation probability.
- **Class-specific Poisson arrivals.** $M_1^D \sim \text{Poisson}(\lambda_1)$ and $M_2^D \sim \text{Poisson}(\lambda_2)$ independently each day.
- **Arrivals pooled and randomly permuted.** All arrivals are combined and randomly shuffled into a single FCFS queue. No class is given scheduling priority.
- **Earliest available slot offered.** Each patient is offered the smallest $r \in \{1, \dots, H - 1\}$ with remaining capacity. The offered delay τ equals this residual day r .
- **Offered delay τ recorded before balking.** The value τ is logged at the time of the offer, regardless of whether the patient subsequently accepts or balks.

- **Balking depends on τ .** A class- i patient accepts with probability $1 - b_i(\tau)$, where $b_i(\cdot)$ is a step function with higher rejection above a class-specific threshold. A balked patient leaves permanently.
- **No-show depends on original τ .** No-show probability $\xi_i(\tau)$ is anchored to the *original* offered delay, not the remaining wait on the day of service.
- **Future cancellations only.** At the start of each day, each booked patient on a future residual day ($r \geq 1$) cancels independently with probability ϕ_i . Canceled slots are returned to the pool and can be rebooked.
- **No same-day cancellations.** Patients at $r = 0$ either attend or are no-shows.
- **No rebooking of no-show slots.** No-show slots count as lost capacity.
- **No-offer rule when horizon is full.** A patient with no slot available is recorded as `no_offer` and excluded from the offered-wait distribution.
- **Utilization counts completed visits per slot.** $\frac{1}{D} \sum_d \frac{\text{served slots}_d}{S}$. No-shows do not contribute.

3 Baseline Diagnosis

3.1 Baseline Parameter Setup

- Two symmetric classes: $\lambda_1 = \lambda_2 = 50$ per day, total $\lambda = 100$.
- $S = 32$ slots per day; $H = 14$ -day booking horizon.
- Cancellation probability: $\phi_1 = \phi_2 = 0.10$.
- Balking threshold: 9 days; high-delay balking probability: 0.50; low: 0.00 (step = 0.50).
- No-show threshold: 6 days; high-delay no-show probability: 0.30; low: 0.00 (step = 0.30).

3.2 Baseline Metric Table

Metric	Baseline	Scenario 2
Average utilization	0.839	1.000
Overall served rate	0.269	0.395
Mean accepted wait (days)	8.35	4.29
Mean offered wait (days)	9.30	5.06
Class served-rate gap (Δ_{access})	0.001	-0.008
Class delay gap (Δ_{delay})	0.003	0.005

Table 1: Baseline and Scenario 2 summary metrics. Scenario 2 changes several assumptions simultaneously and is a comparison point, not a one-parameter causal test.

3.3 Utilization–Access Gap

$$\text{average utilization} = \frac{\text{completed visits}}{\text{available slots}}, \quad \text{overall served rate} = \frac{\text{completed visits}}{\text{arrivals}}.$$

3.4 Demand-to-Capacity Calculation

With $\lambda = 100$ and $S = 32$, the demand-to-capacity ratio is $100/32 \approx 3.1$. Even with zero no-shows and zero cancellations, the served rate is bounded by $32/100 = 0.32$. The observed 0.269 falls below this ceiling because no-shows eliminate completed visits after scheduling.

4 Arrival Outcome Decomposition

Outcome	Share	Mechanism
Served	0.269	Accepted, did not cancel, attended
Balked	0.356	Offered a slot; rejected due to long delay
Canceled	0.323	Accepted; canceled before service day
No-show	0.052	Accepted; not canceled; did not attend
No-offer	0.000	Horizon full; no slot offered
Total	≈ 1.000	

Table 2: Arrival outcome decomposition at baseline.

Baseline is not horizon-saturated. No-offer is zero, so access failure is not caused by the horizon filling.

Low served rate is driven by balking and cancellation. Together they account for $0.356 + 0.323 = 0.679$ of arrivals. Reducing no-shows alone would not materially raise the served rate.

High cancellation is mechanically plausible. A patient booked 10 days ahead faces about 10 independent cancellation draws. With $\phi = 0.10$, the survival probability is $(0.90)^{10} \approx 0.35$. This should be verified in code to confirm cancellation is applied *once per residual day per patient*, not once per booking.

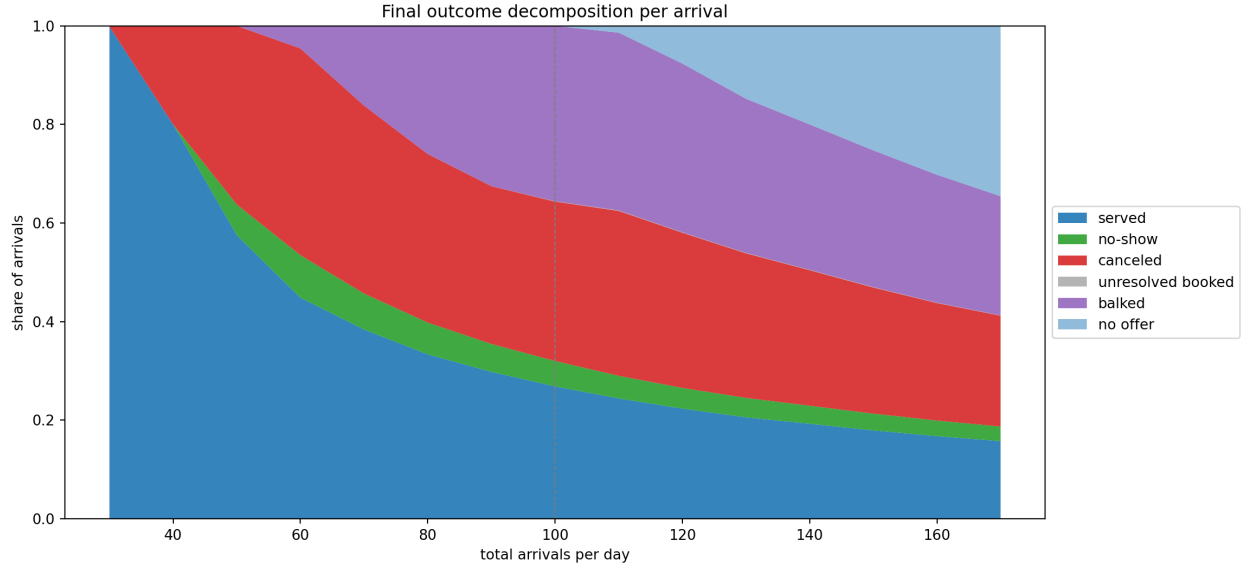


Figure 1: Arrival outcome decomposition under demand pressure. At baseline ($\lambda = 100$), no-offer is zero. No-offer becomes substantial only above approximately $\lambda = 110$. Balking and cancellation dominate at baseline.

5 Cross-Metric Diagnostics

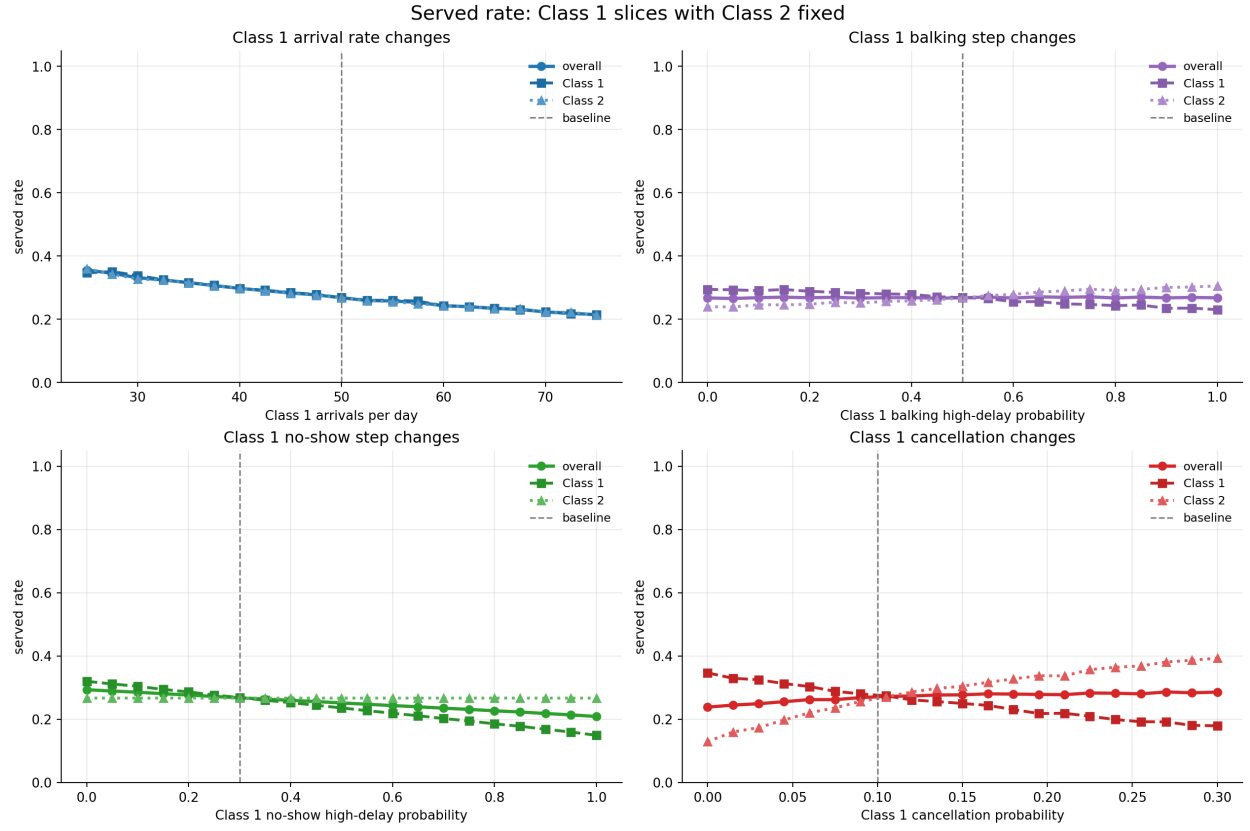


Figure 2: Overall served rate as demand changes. Class 1 varies on the x-axis while Class 2 stays fixed at baseline.

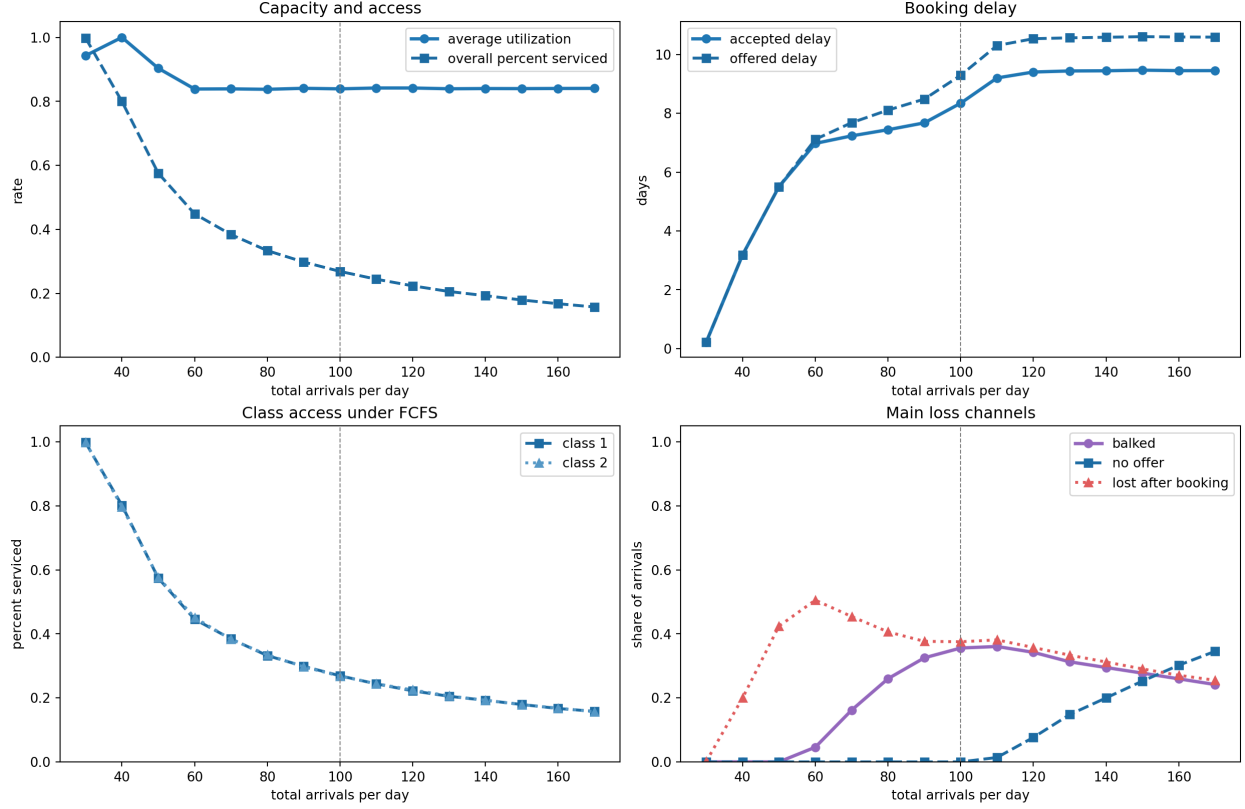


Figure 3: FCFS capacity stress curves ($\lambda = 30$ to 170, 50/50 mix, three seeds averaged). Utilization stays near 0.84 across a wide demand range while served rate falls steeply. No-offer becomes material only above approximately $\lambda = 110$.

Three diagnostic patterns emerge.

- 1. Utilization can stay near 0.84 while served rate falls.** Slots are still occupied by patients who accept; completed visits per slot stays high even as the served fraction of arrivals collapses.
- 2. Accepted wait is selected downward by balking.** Patients who would have waited longest have left the system. Offered wait includes those rejected patients and is the less biased measure.
- 3. No-offer becomes important only above $\lambda \approx 110$.** At baseline demand no-offer is zero. The no-offer share must be reported separately whenever it is non-negligible.

6 Metric Hierarchy Under This Formulation

Metric	What it measures	Interpretation	Main caveat
Overall served rate	Fraction of arrivals completing a visit	Primary access metric	Bounded above by S/λ
Average utilization	Fraction of slots filled by completed visits	Capacity-use metric	Not an access metric
Mean offered wait	Avg delay offered to all patients receiving an offer	Broader congestion measure	Excludes no-offer patients
Mean accepted wait	Avg delay among patients who booked	Patient-level delay for those who accepted	Selected downward by balking
Balking rate	Fraction of offered patients rejecting the delay	Diagnostic for rejected offers	Not a success metric
Class served-rate gap	Class 1 served rate minus Class 2	Class-level access disparities	Near zero under symmetric parameters

Table 3: Metric hierarchy under the current FCFS formulation.

7 Sensitivity Screen: Main Patterns

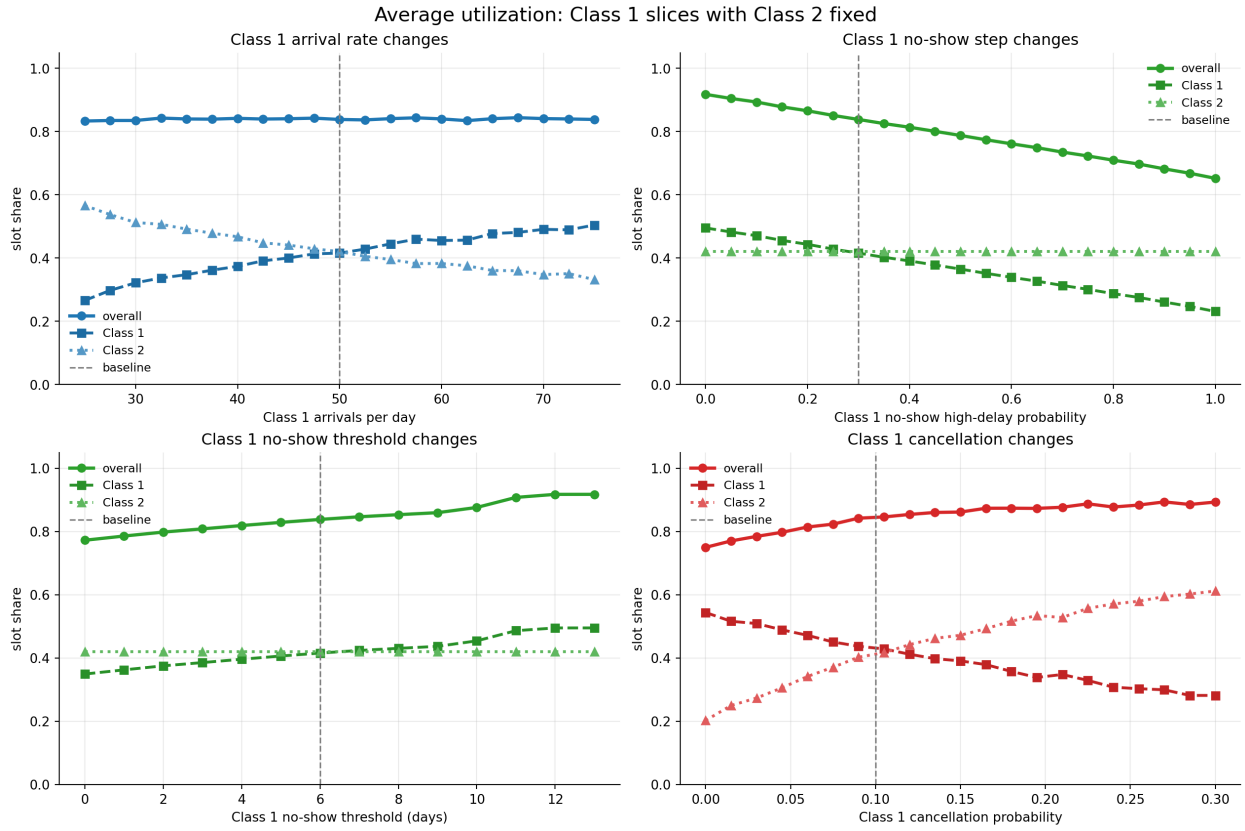


Figure 4: Average utilization driver plots. No-show behavior is the clearest utilization driver.

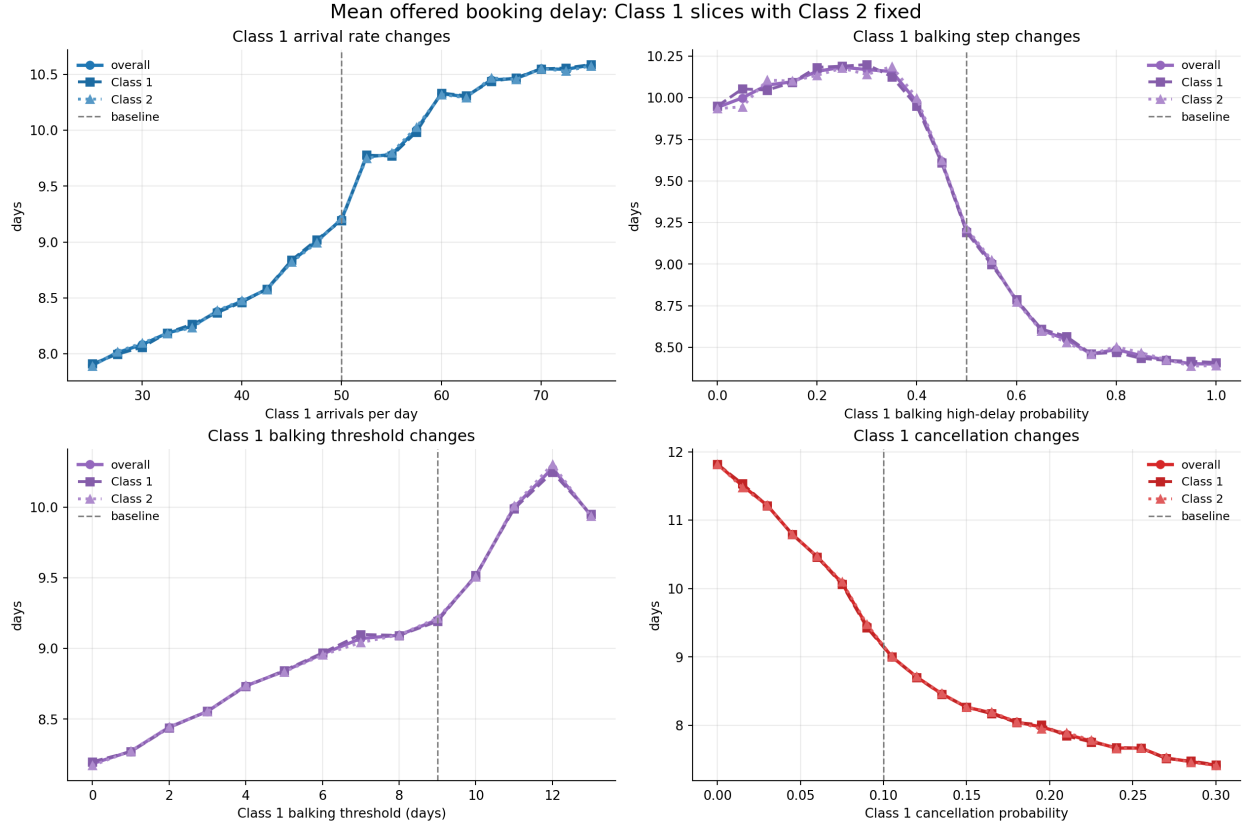


Figure 5: Mean offered wait driver plots. Arrival pressure is the strongest driver. Higher balking or cancellation can shorten offered wait for reasons that worsen access rather than improve it.

Demand pressure raises offered wait and reduces served rate monotonically. Utilization is relatively insensitive across a wide demand range.

No-show behavior is the primary utilization driver. The regression screen (Appendix ??) identifies the no-show threshold as the strongest single predictor of utilization.

Cancellation has ambiguous aggregate effects. Cancelled slots are returned to the pool and can be rebooked, so higher cancellation can shorten offered wait while still reducing completed visits per original patient. The regression screen shows a *positive* coefficient for cancellation on both utilization and served rate, which is counterintuitive and should be verified at code level (Section ??).

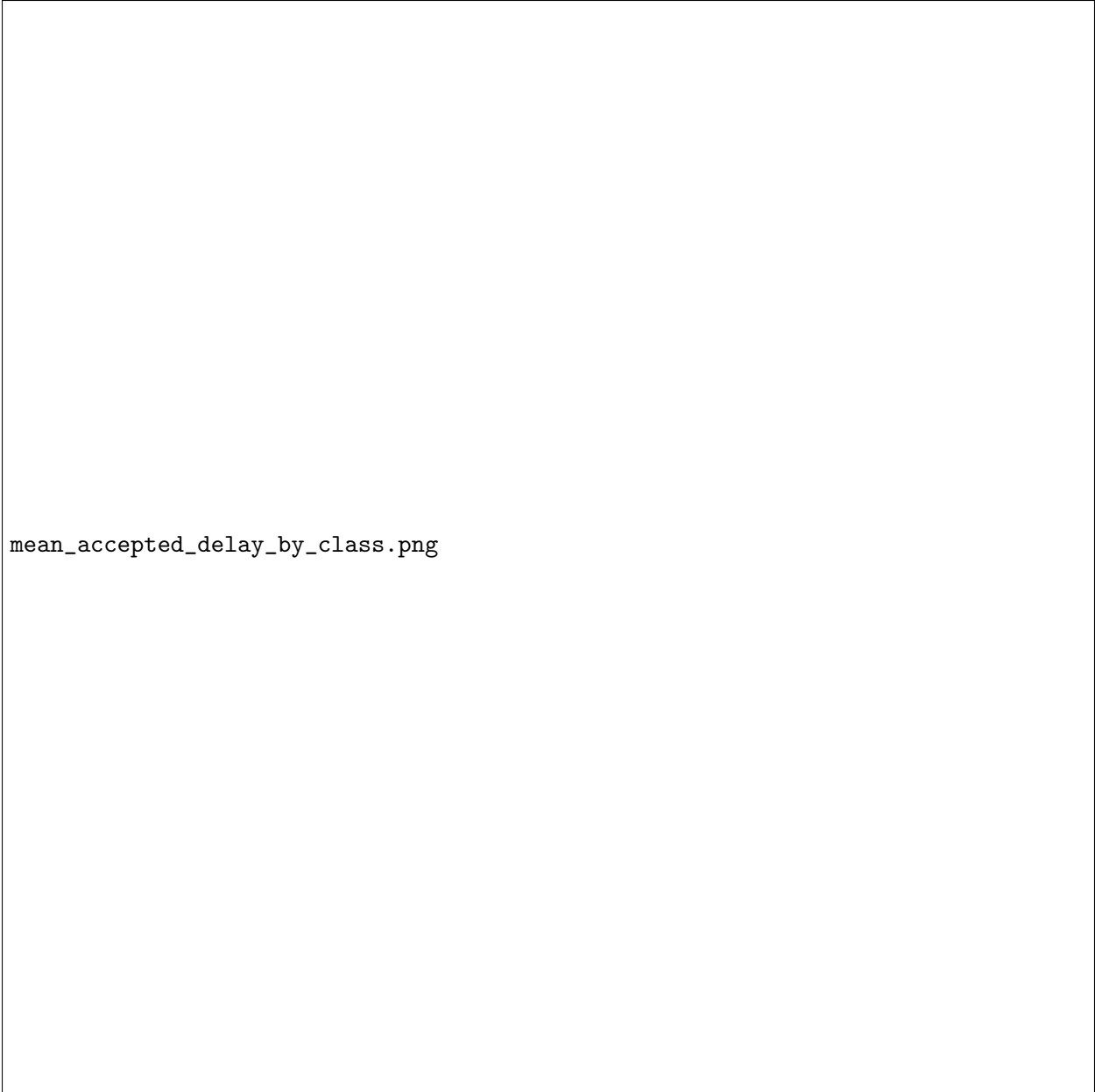
Balking can reduce accepted wait while reducing access. This is the main argument for reporting offered wait rather than accepted wait as the primary delay metric.

Class gaps arise from asymmetric behavioral parameters. Cancellation probability differences are the largest class-access driver in the regression screen.

8 Balking Deep Dive: Selection vs. Access



Figure 6: Class 1 high-balking-probability sweep: served rate by class. As Class 1 rejection probability rises, Class 1 access falls while Class 2 access rises slightly.



mean_accepted_delay_by_class.png

Figure 7: Class 1 high-balking-probability sweep: mean accepted delay by class. Class 1 accepted delay falls as balking rises because long-delay patients exit the accepted sample. This is a selection effect.

Main point. Increasing balking can make accepted wait appear to improve while access worsens. Offered wait and served rate together are necessary to interpret the change correctly.

9 Class-Gap Interpretation Under FCFS

Under the current formulation, class labels are *exchangeable* when parameters are symmetric. Arrivals are pooled and randomly permuted; no class receives FCFS priority.

Symmetric parameters imply expected class gap near zero. The observed baseline gap of 0.001 is consistent with simulation noise.

Gap drivers in order of absolute standardized coefficient (Appendix ??):

1. Cancellation probability gap: $\hat{\beta} = -0.452$
2. Balking threshold gap: $\hat{\beta} = +0.429$
3. Balking step gap: $\hat{\beta} = -0.349$
4. No-show threshold gap: $\hat{\beta} = +0.268$
5. No-show step gap: $\hat{\beta} = -0.211$
6. Class 1 arrival share: $\hat{\beta} = -0.120$

10 What Seems Mechanically Robust vs. What Needs Validation

10.1 Mechanically Robust

- Utilization and served rate have different denominators. High utilization can coexist with low served rate by definition.
- The demand ceiling: served rate cannot exceed $32/100 = 0.32$ without overbooking.
- Offered wait is less selected than accepted wait.
- No-offer is a distinct access failure excluded from wait metrics.
- Class labels are exchangeable under symmetric assumptions.
- No-offer is zero at baseline; failures are behavioral, not capacity.

10.2 Needs Validation or Calibration

- **Cancellation applied once per residual day.** Verify in code: one draw per patient per day, not one draw per booking.
- **Positive cancellation coefficient on served rate.** Counterintuitive; verify what fraction of canceled slots are rebooked.
- **No-offer propagation within a day.** Confirm all remaining queue patients receive no-offer when horizon is full; confirm permutation is re-drawn each day.
- **No-show slots never rebooked.** Deliberate choice; should be stated explicitly when comparing to literature models.
- **Same-day cancellations excluded.** All same-day losses route through no-show.
- **Class-gap stability with more seeds.** Current grids use one seed per grid point.

- **Realism of the demand/capacity ratio.** 3.1 arrivals per slot is a stress setting, not a calibrated clinic baseline.
- **Behavioral parameter calibration.** Balking step 0.50, no-show step 0.30, and cancellation 0.10 are assumed, not estimated.

11 Summary for Discussion

The simulation output suggests the following under the current formulation.

Utilization near 0.84 and served rate near 0.27 can coexist because they measure different things. The structural ratio $\lambda/S \approx 3.1$ constrains served rate below 0.32 even before behavioral losses. Balking and cancellation reduce it further and dominate no-show in the baseline decomposition.

The metric ordering is: (1) overall served rate for access, (2) average utilization for slot use, (3) mean offered wait for patient-facing delay, (4) no-offer share as a separate indicator, (5) mean accepted wait as a secondary selected metric, (6) balking rate and class gap as diagnostics.

The sensitivity screen is mechanically consistent with this ordering. Several findings should be checked before outputs support design decisions.

Questions for discussion:

1. Is the baseline demand/capacity ratio intentional as a stress scenario, or should it be recalibrated?
2. Is the positive cancellation coefficient verified at code level? What fraction of canceled slots are rebooked?
3. Should same-day cancellations be allowed?
4. Should no-show slots be rebooked?
5. Are threshold-jump step functions the right behavioral model, or should smoother functions be explored?
6. Are the class behavioral parameters population-specific or sensitivity levers only?

A Exact Metric Definitions and Notation

Let $i \in \{1, 2\}$, D = measured days, S = slots/day, H = horizon length. For class i : A_i arrivals, B_i booked, K_i balked, $O_i = B_i + K_i$ offered, N_i no-offer, C_i canceled, H_i no-shows, V_i served. $A_i = V_i + C_i + H_i + K_i + N_i$.

$$\begin{aligned}
\text{percent_serviced}_i &= V_i/A_i, \\
\text{overall_percent_serviced} &= \sum_i V_i / \sum_i A_i, \\
\text{average_utilization} &= \frac{1}{D} \sum_d \text{served slots}_d / S, \\
\text{mean_offered_booking_delay} &= \sum_i \text{total offered delay}_i / \sum_i O_i, \\
\text{mean_accepted_booking_delay} &= \sum_i \text{total accepted delay}_i / \sum_i B_i, \\
\text{balking_rate} &= \sum_i K_i / \sum_i O_i, \\
\Delta_{\text{access}} &= \text{percent_serviced}_1 - \text{percent_serviced}_2, \\
\Delta_{\text{delay}} &= \bar{\tau}_2^{\text{offered}} - \bar{\tau}_1^{\text{offered}}, \\
\Delta_{\text{balking}} &= K_1/O_1 - K_2/O_2.
\end{aligned}$$

Behavioral step functions:

$$b_i(\tau) = \begin{cases} b_i^{\text{low}} & \tau < \theta_i^{\text{balk}}, \\ b_i^{\text{high}} & \tau \geq \theta_i^{\text{balk}}. \end{cases}$$

Balking step = $b_i^{\text{high}} - b_i^{\text{low}}$. No-show $\xi_i(\tau)$ has the same structure.

B Sensitivity Design

All grids centered on `configs/baseline.yaml`. One-axis slices vary Class 1 while Class 2 stays at baseline. Two-class heatmaps vary both classes independently. Each grid point uses seed 2027; multi-seed comparisons average seeds 2027, 2028, 2029.

Parameter	Baseline	Range	Grid size
Arrivals (per class), 1D slices	50	25–75	continuous
Total arrivals, FCFS stress	100	30–170	continuous
Balking step	0.50	0.00–1.00	21
Balking threshold (days)	9	0–13	14
No-show step	0.30	0.00–1.00	21
No-show threshold (days)	6	0–13	14
Cancellation probability	0.10	0.00–0.30	21

Table 4: Sensitivity parameter ranges.

Regression screen: 240 randomized settings (seed 9137); two seeds (4101, 4102) averaged per setting; 12 features (total arrival rate, Class 1 share, average level and Class 1–Class 2 gap for each behavioral parameter); OLS with HC3 robust SEs; only $p < 0.05$ coefficients shown.

Balking deep dives: Class 1 high-balking-probability sweep over $\{0.0, 0.1, \dots, 0.9\}$; Class 1 threshold sweep over $\{3, \dots, 13\}$; 100 seeds per point; Class 2 fixed at baseline.

C Supplementary Sensitivity Plots

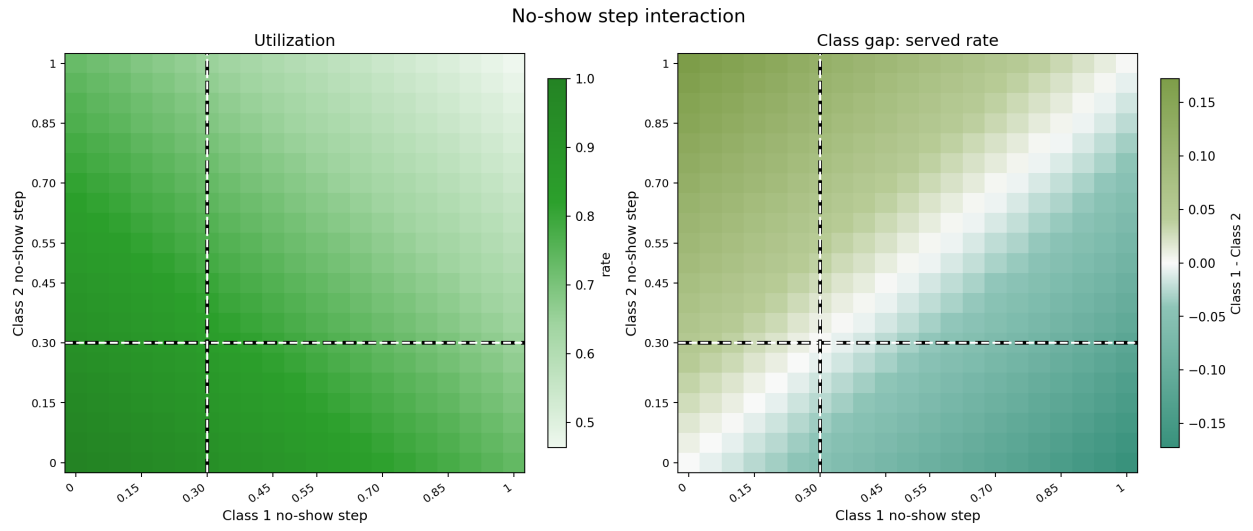


Figure 8: No-show step heatmap. Left: utilization. Right: Class 1 served-rate gap.

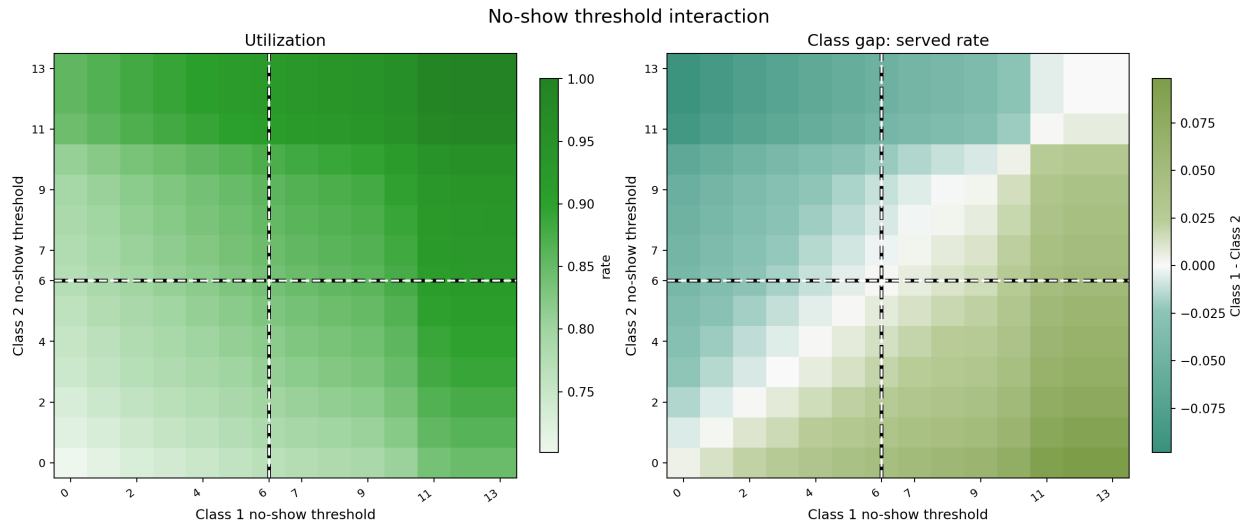


Figure 9: No-show threshold heatmap. Left: utilization. Right: Class 1 gap.

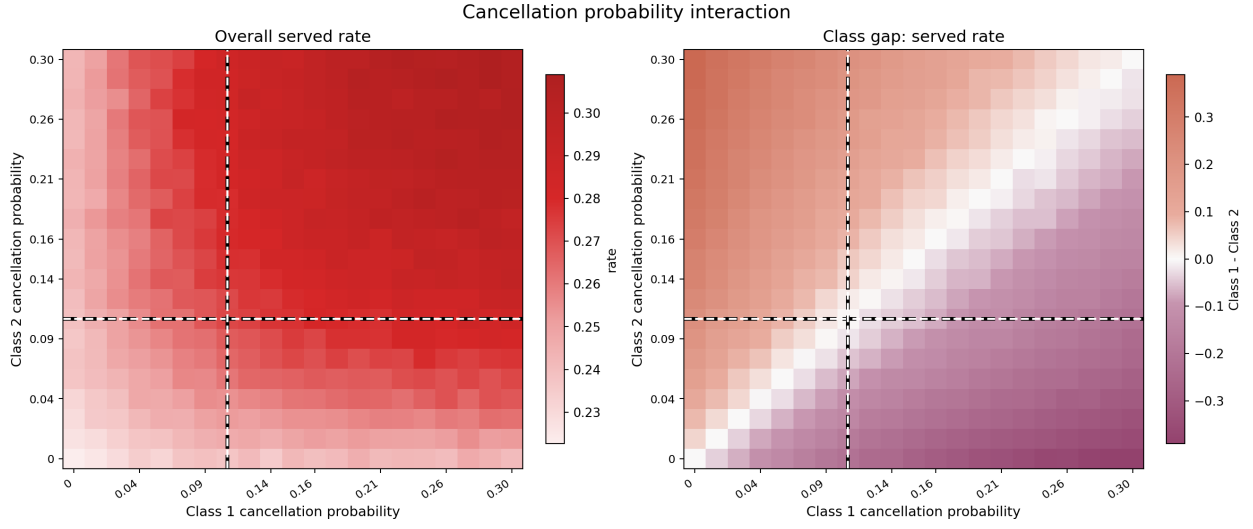


Figure 10: Cancellation heatmap. Left: served rate. Right: Class 1 gap.

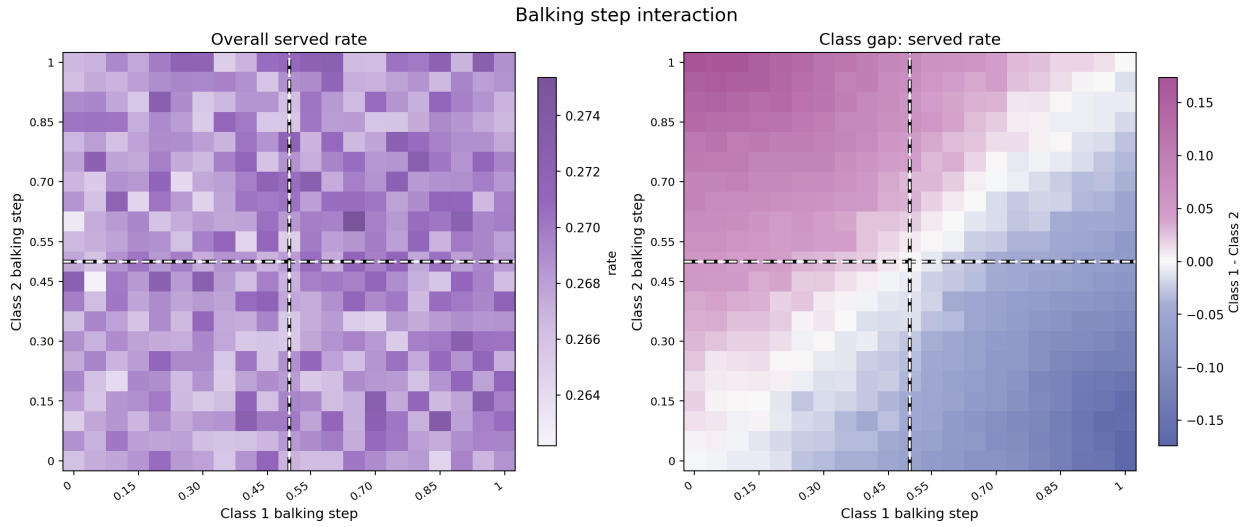


Figure 11: Balking step heatmap. Left: served rate. Right: Class 1 gap.

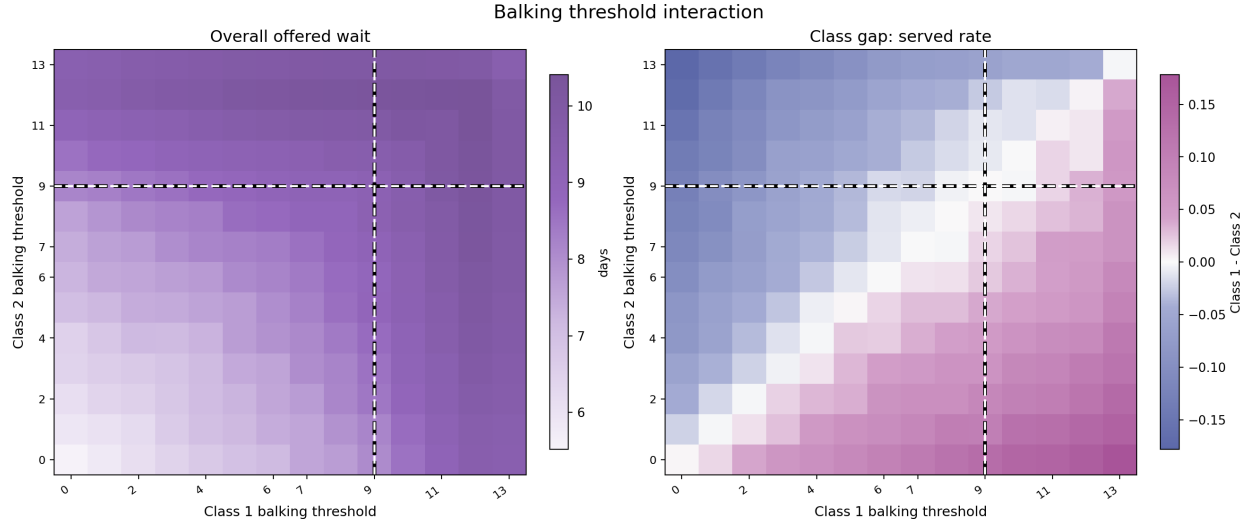


Figure 12: Balking threshold heatmap. Left: offered wait. Right: Class 1 gap.

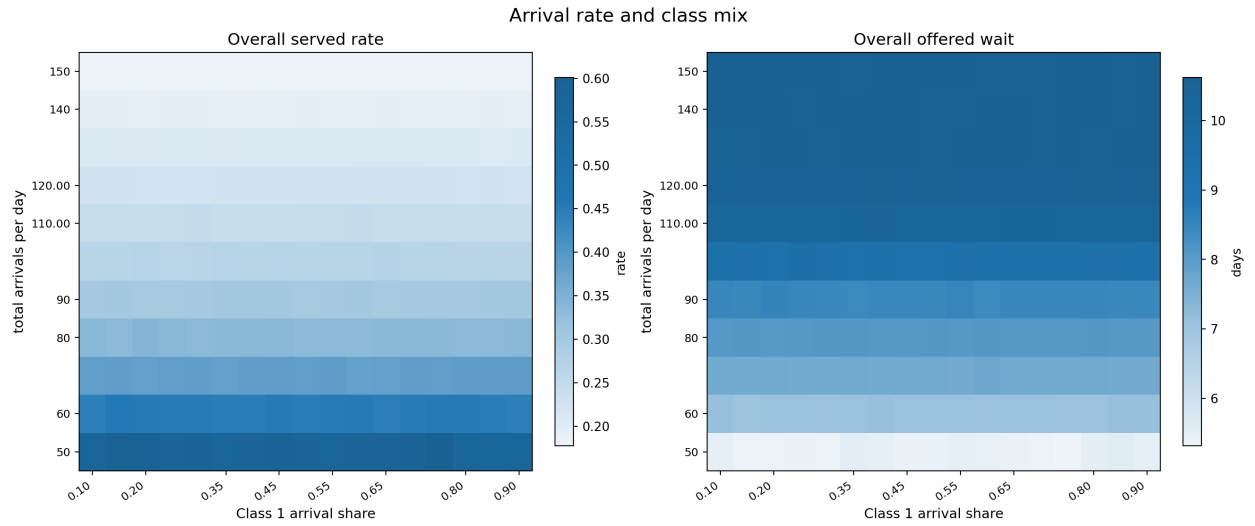


Figure 13: Arrival mix heatmap. Left: served rate. Right: offered wait.

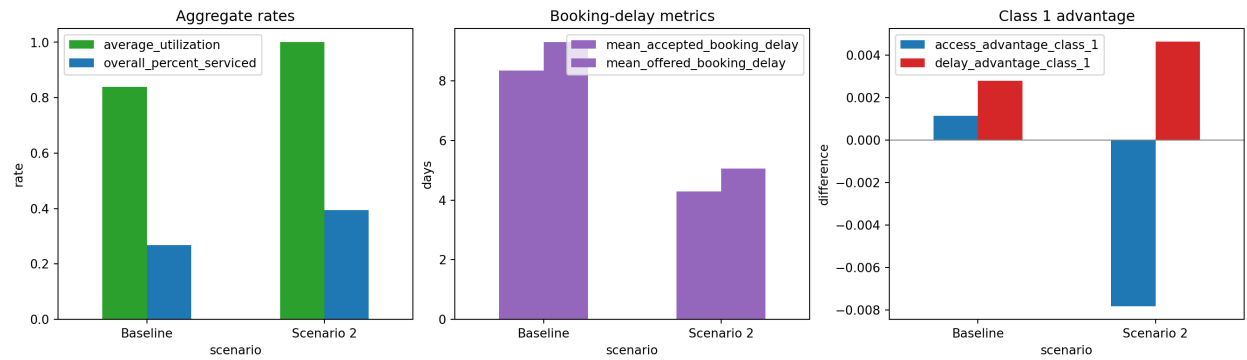


Figure 14: Scenario-level metric comparison (baseline vs. Scenario 2).

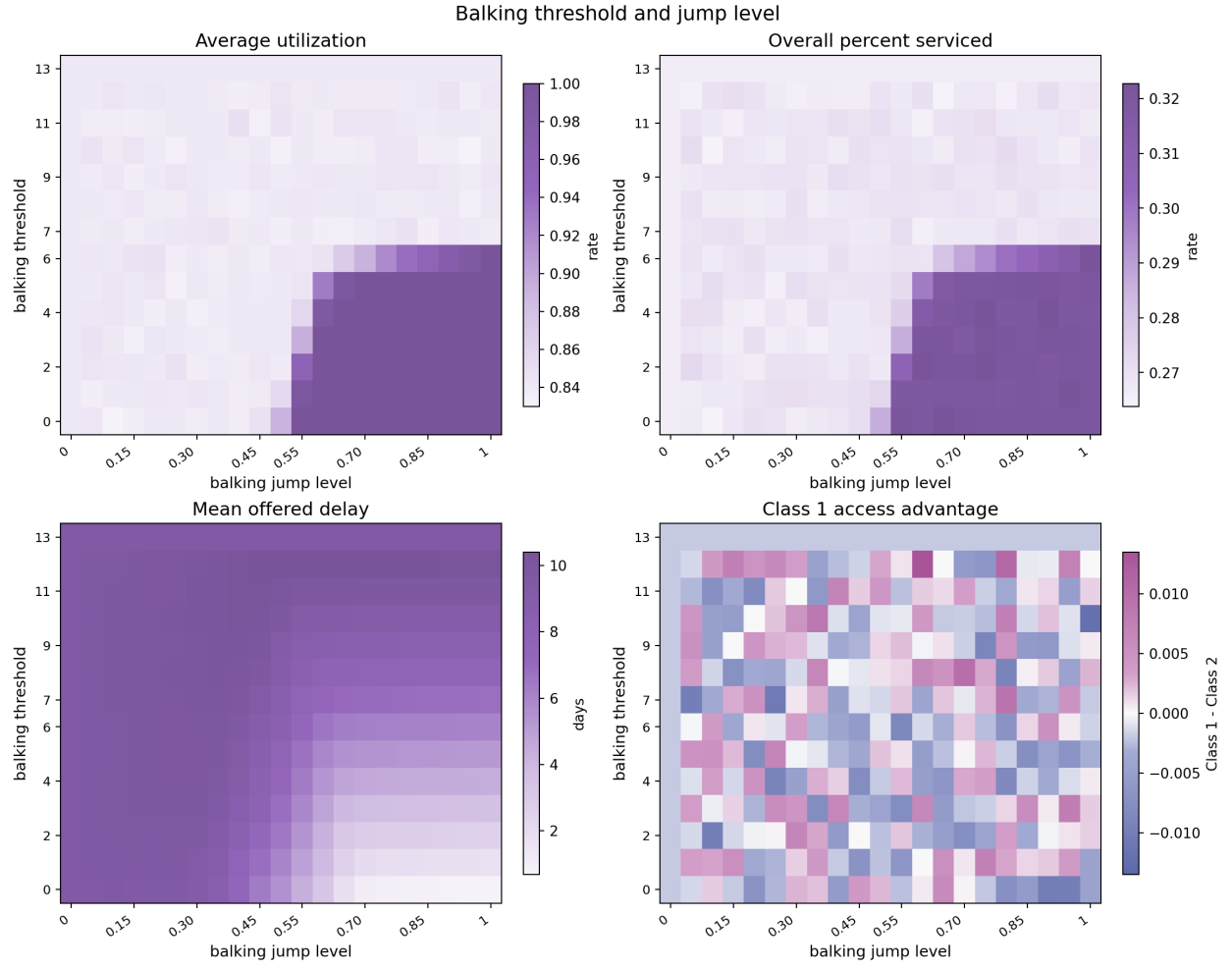


Figure 15: Common balking threshold and jump level.

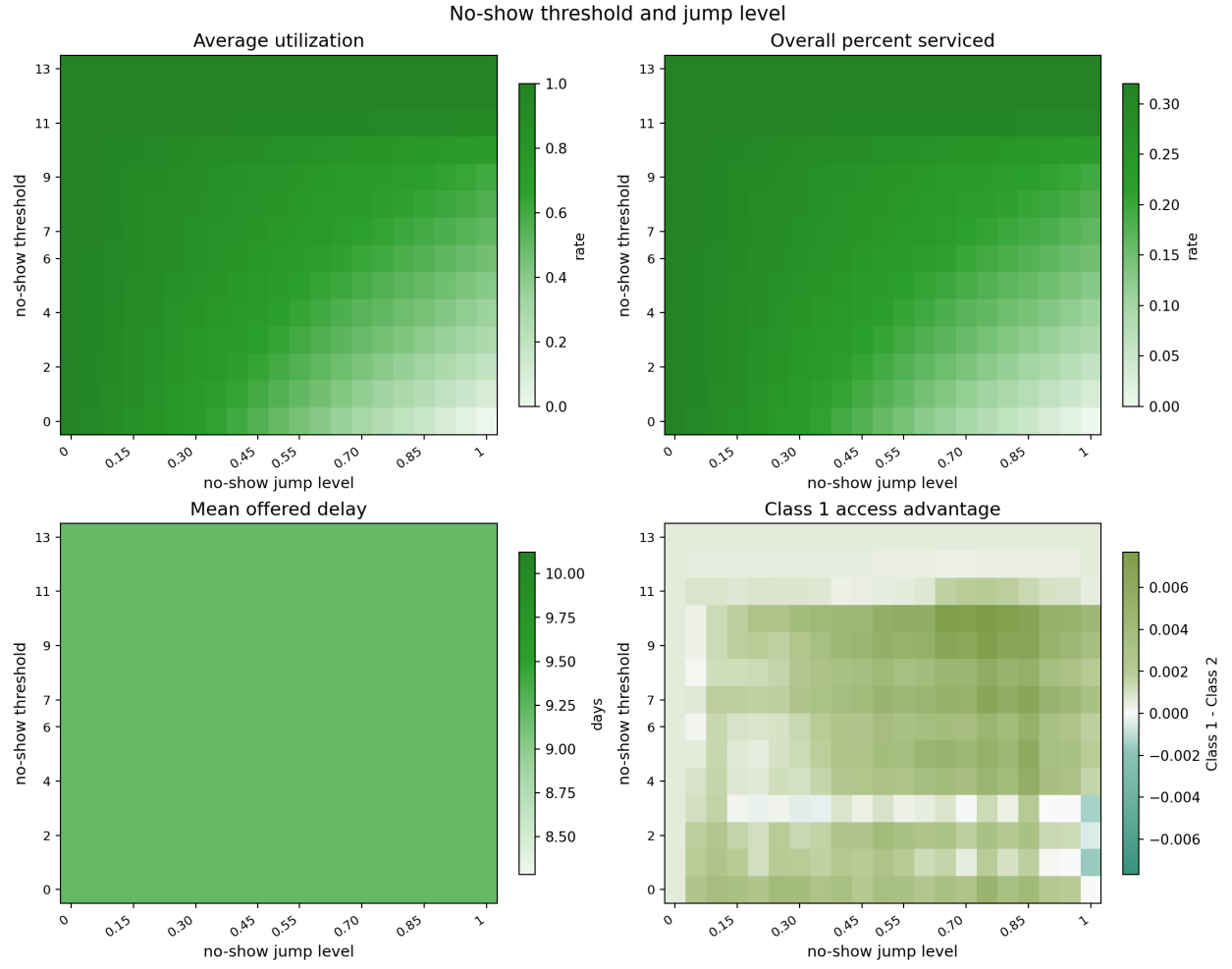


Figure 16: Common no-show threshold and jump level.

D Regression Screen

Target	Feature	Std. coef.	95% HC3 CI	HC3 p
Utilization	Avg no-show threshold	+0.512	[0.433, 0.591]	< 0.001
Utilization	Avg no-show step	-0.423	[-0.501, -0.346]	< 0.001
Utilization	Avg cancellation prob.	+0.320	[0.247, 0.393]	< 0.001
Utilization	Total arrival rate	-0.109	[-0.188, -0.031]	0.006
Utilization	Avg balking threshold	-0.086	[-0.162, -0.009]	0.028
Overall served rate	Total arrival rate	-0.774	[-0.855, -0.694]	< 0.001
Overall served rate	Avg no-show threshold	+0.251	[0.193, 0.308]	< 0.001
Overall served rate	Avg no-show step	-0.175	[-0.232, -0.119]	< 0.001
Overall served rate	Avg cancellation prob.	+0.117	[0.058, 0.176]	< 0.001
Offered wait	Total arrival rate	+0.576	[0.499, 0.653]	< 0.001
Offered wait	Avg cancellation prob.	-0.441	[-0.506, -0.376]	< 0.001
Offered wait	Avg balking threshold	+0.278	[0.202, 0.353]	< 0.001
Offered wait	Avg balking step	-0.206	[-0.291, -0.122]	< 0.001
Class gap	Cancellation prob. gap	-0.452	[-0.558, -0.347]	< 0.001
Class gap	Balking threshold gap	+0.429	[0.331, 0.528]	< 0.001
Class gap	Balking step gap	-0.349	[-0.431, -0.266]	< 0.001
Class gap	No-show threshold gap	+0.268	[0.179, 0.357]	< 0.001
Class gap	No-show step gap	-0.211	[-0.286, -0.136]	< 0.001
Class gap	Class 1 arrival share	-0.120	[-0.209, -0.030]	0.009

Table 5: Significant standardized regression coefficients ($p < 0.05$, HC3 robust SEs). Gap = Class 1 minus Class 2 value.

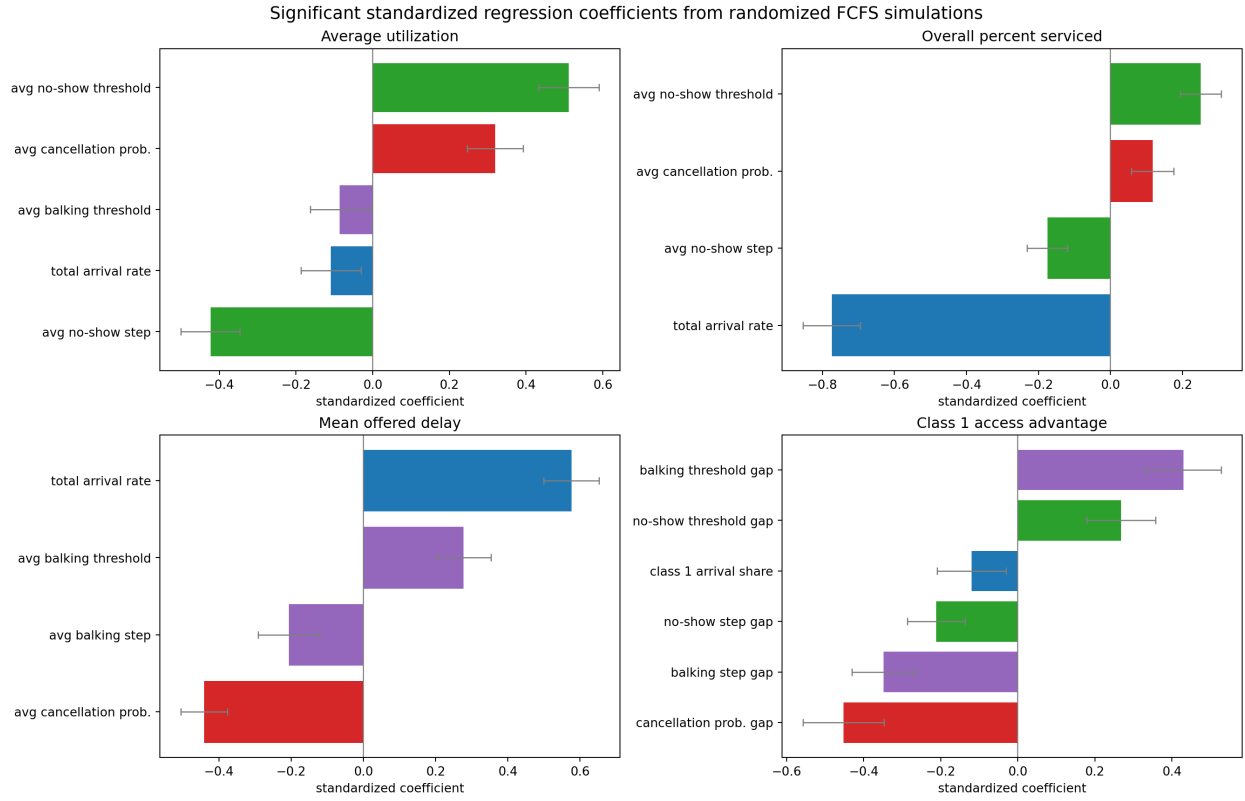


Figure 17: Significant standardized regression coefficients with 95% HC3 confidence intervals.